# NYC Airbnb Analysis: Ratings, Pricing, and Room Type Insights

**Course:** Data Storytelling and Visualization
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**Abstract**

***This report investigates Airbnb listings across New York City, uncovering how pricing, availability, and guest satisfaction metrics vary across neighborhoods and property types. Using Tableau dashboards, we conducted dynamic metric analysis and visual exploration to identify customer behavior patterns. Key takeaways reveal that service quality—especially cleanliness—has a greater impact on satisfaction than price, and private rooms consistently offer the best value in terms of affordability and guest ratings.***

# 1 Introduction

As Airbnb reshapes the short-term rental market in urban hubs like NYC, understanding customer preferences and listing trends becomes crucial. This project explores public Airbnb datasets through interactive Tableau dashboards to derive actionable business insights related to price, reviews, and occupancy.

# 2 Research Objectives

The analysis was guided by the following questions:

- How do price, availability, and reviews vary across NYC neighborhoods?
- Does higher price correlate with better reviews?
- Which review factors (e.g., cleanliness, communication) matter most?
- Which room/property types provide the best guest experiences?
- Where should hosts invest or adjust their strategies?

# 3 Methodology

This project employed a structured, data-driven approach to analyze Airbnb listings across New York City using public datasets and Tableau for visualization. The methodology was divided into five main phases:

## 3.1. Data Collection

We sourced the dataset from *Inside Airbnb* (2023), which provides publicly available listings data for major cities. The NYC dataset includes detailed fields such as listing ID, price, number of reviews, availability, neighborhood, room type, and review scores across multiple dimensions (e.g., cleanliness, communication).

## 3.2. Data Cleaning and Preprocessing

Using Tableau's in-built data preparation tools, we performed the following steps:

- Removed duplicate records and empty/null entries.
- Filtered out listings with zero availability or zero reviews to ensure data relevance.
- Normalized metrics such as price and availability to enable fair comparison across neighborhoods and room types.
- Converted review metrics from multiple categorical dimensions into numerically usable format.

## 3.3. Exploratory Data Analysis (EDA)

We conducted EDA in Tableau to identify trends, outliers, and distribution patterns. Key observations at this stage informed the design of our dashboards. For instance, we observed that:

- Entire home listings dominated the dataset.
- Price varied significantly by neighborhood and room type.
- Some boroughs like Staten Island consistently ranked high in review metrics.

## 3.4. Dashboard Design and Visualization

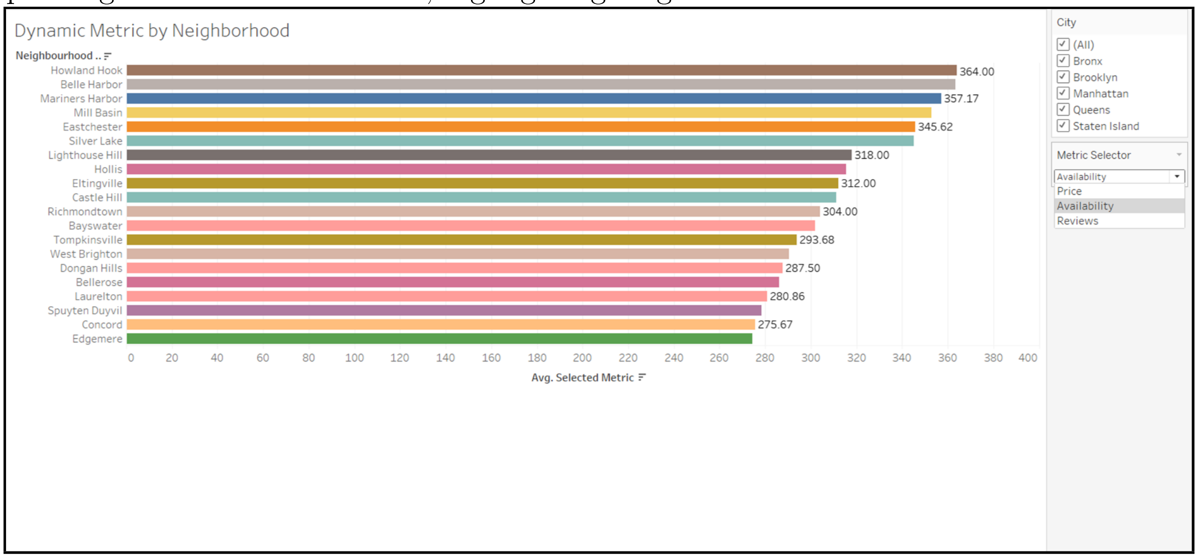
Two dynamic dashboards were created using Tableau:

- **Dashboard 1 – NYC Airbnb Trends:** Focused on listing share, booking growth, price trends, and availability.
- **Dashboard 2 – Market Performance:** Highlighted top neighborhoods, review ratings by room/property type, and cleanliness scores by city.

# 4 Findings and Insights

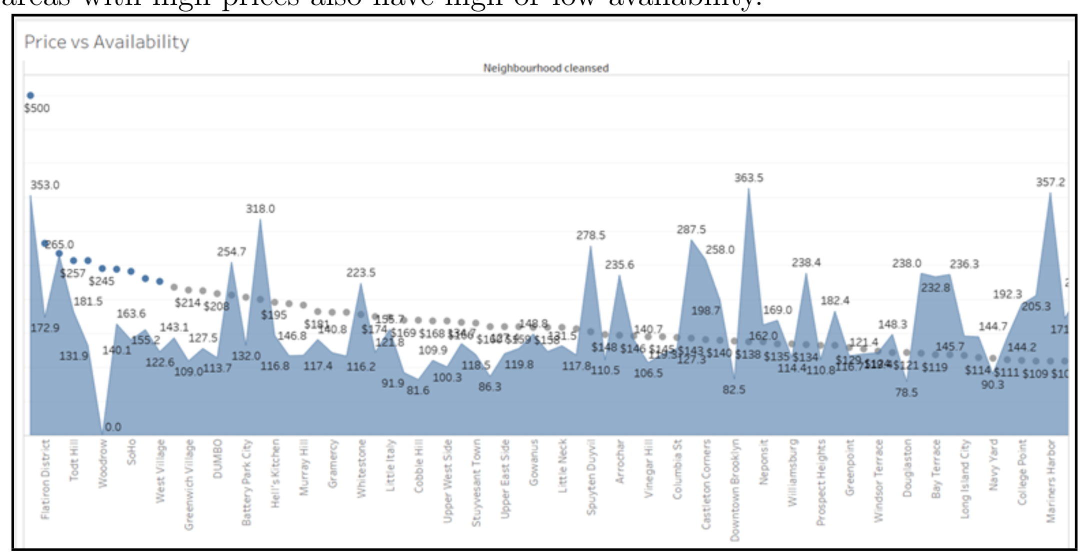
## 4.1 Dynamic Metric by Neighborhood

Figure 1: This chart dynamically shows variations in average price, availability, or reviews depending on the selected metric, highlighting neighborhood differences.



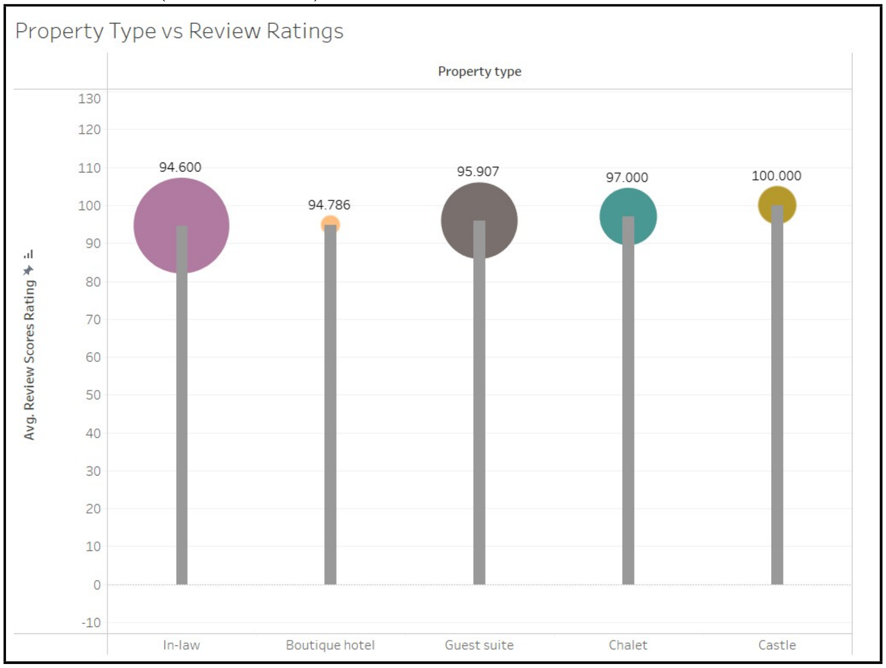

## 4.2 Price vs Availability of Neighborhood

Figure 2: This dual-axis chart compares price and availability across neighborhoods, revealing if areas with high prices also have high or low availability.



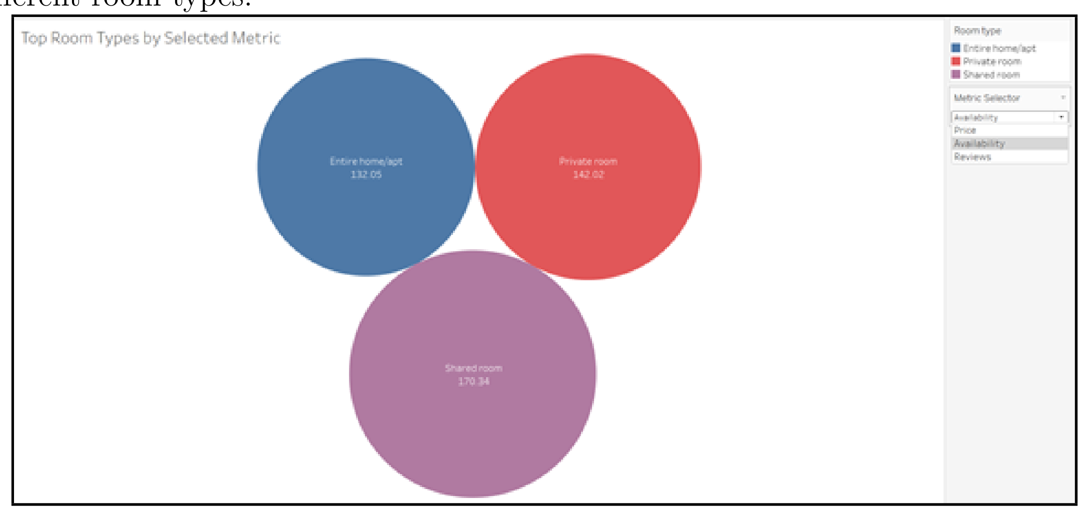
## 4.3 Property Type vs Review Ratings

Figure 3: By comparing review ratings across different property types, you can observe if higher-priced properties (like castles) are rated better than lower-priced ones.



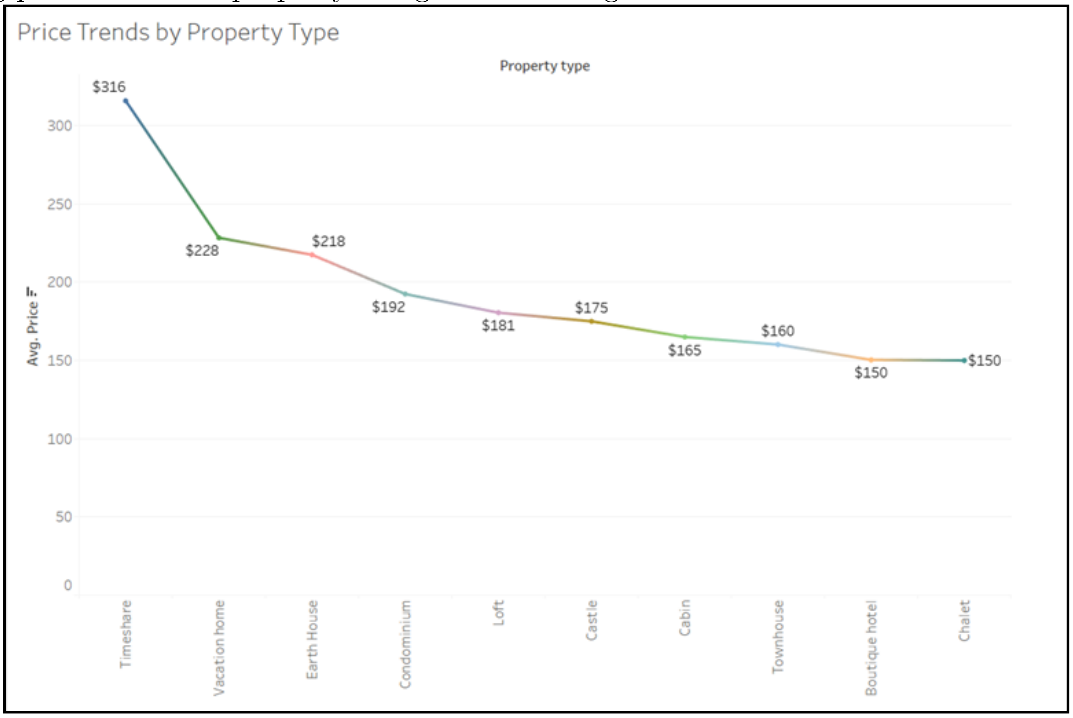

## 4.4 Top Room Types by Selected Metric

Figure 4: When selecting "Reviews" in the Metric Selector, the size of the bubbles will adjust depending on the selected metric, making it easier to compare the average review metrics for different room types.



## 4.5 Price Trends by Property Type

Figure 5: This chart shows how average prices differ based on property type, helping analyze pricing patterns across property categories and neighborhoods.




## 4.6 Selected Review Score by City

Figure 6: This bar chart allows you to dynamically switch between review types to see which aspects vary most across cities and which score matters for highly-rated listings.

# 5 Conclusion

From an analyst's perspective, the dashboards reveal that service quality—particularly cleanliness—plays a pivotal role in guest satisfaction across NYC Airbnb listings. Pricing does not directly correlate with better availability or reviews, and private rooms consistently deliver high value. Certain boroughs like Staten Island outperform others in review metrics, while niche property types like Castles and Chalets score exceptionally well. The findings strongly support data-driven decisions to optimize property type offerings, improve host practices, and guide Airbnb's strategic expansion.

# 6 Recommendations

Airbnb should focus on optimizing pricing in underutilized neighborhoods, promote high-performing property types like private rooms and chalets, and invest in improving key service areas such as cleanliness and communication to boost guest satisfaction and occupancy rates.

1. Focus marketing in high-availability, low-saturation neighborhoods like Howland Hook.
2. Promote private rooms—they provide high satisfaction at lower prices.
3. Improve host education around cleanliness and communication to raise review scores.
4. Explore niche high-performing property types (e.g., Chalets, Castles).

# 7 Future Scope

This project showcases how interactive data visualization can uncover actionable insights into Airbnb's pricing, availability, and guest preferences, enabling data-driven decisions to optimize listings and improve customer satisfaction.

- Extend the analysis to post-pandemic booking patterns to evaluate shifts in guest behavior.
- Integrate predictive analytics to enable dynamic pricing and occupancy forecasting.
- Benchmark New York City performance against other major cities to identify market strengths and gaps.

# 8 Dashboards

## 1: NYC Airbnb Trends: Pricing, Availability, and Room Type Insights

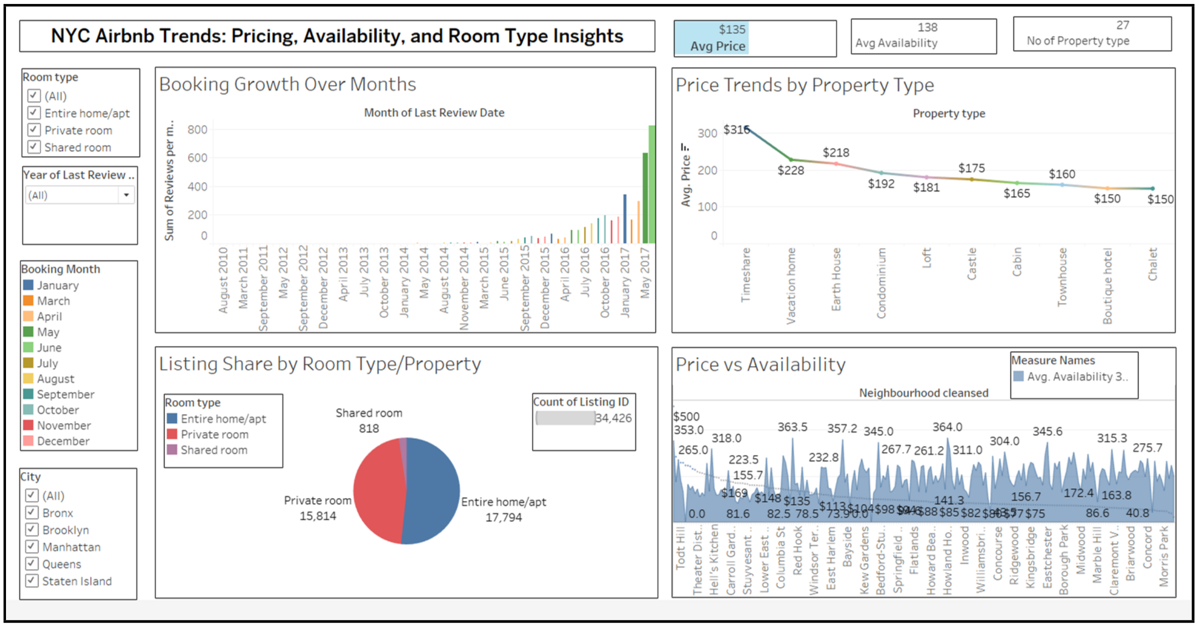



Figure 7: Dashboard 1—NYC Airbnb Trends: Pricing, Availability, and Room Type Insights

### Insights from Dashboard 1:

- Booking activity has steadily increased from 2010 to 2017.
- Entire home/apartments dominate listings; shared rooms are the least common.
- Price trends show clear segmentation—Timeshares are premium; Chalets are budget-friendly.
- High availability is often found in lower-priced outer boroughs.

## 2: Dynamic Insights into NYC Airbnb Market Performance

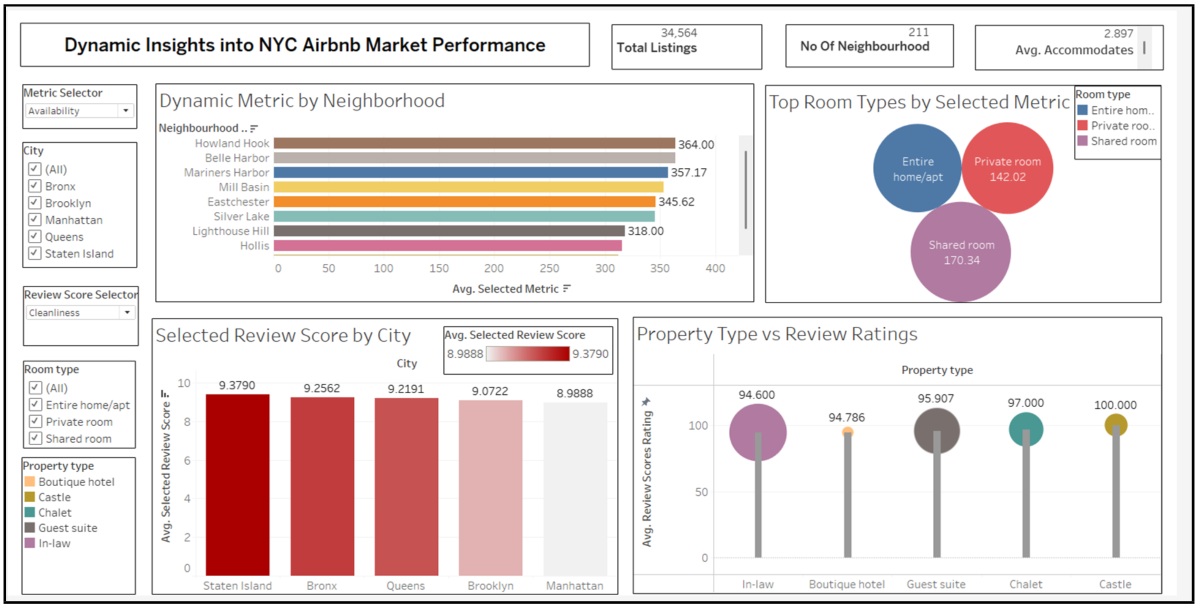



Figure 8: Dashboard 2 – Dynamic Insights into NYC Airbnb Market Performance

### Insights from Dashboard 2:

- Staten Island leads in cleanliness ratings, indicating superior service quality.
- Guest satisfaction is highest for niche property types like Castles and Chalets.
- Private rooms balance price and performance—ideal for budget travelers.
- Howland Hook, Belle Harbor, and Mill Basin are top neighborhoods in selected metrics.

# 9 References


- Airbnb Ratings Dataset — `https://www.kaggle.com/datasets/samyukthamurali/airbnb-ratings-dataset?select=NY_Listings.csv`
- Tableau Public (2024). Airbnb Dashboards — `https://10ay.online.tableau.com/#/site/shashayu/workbooks/2608068?:origin=card_share_link`